

Federated Quantum Machine Learning for Multi-Site Schizophrenia Classification from fMRI Connectomes

Subproject 6: Distributed Brain Connectivity Analysis

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SECTION 1: RESEARCH DEFINITION

1.1 Problem Statement

Schizophrenia (SZ) affects approximately 24 million people globally, yet diagnosis remains entirely symptom-based under DSM-5/ICD-11 criteria, with misdiagnosis rates exceeding 25% in first-episode cases. Resting-state functional MRI (rs-fMRI) has revealed systematic functional connectivity (FC) alterations as candidate biomarkers, with machine learning classifiers achieving 70-96% accuracy in single-site studies (though cross-site prospective validation drops to 77.3%, establishing the realistic clinical baseline). However, clinical translation is obstructed by four compounding barriers: (1) small per-site sample sizes yielding underpowered models; (2) scanner-induced site effects producing substantial domain shift (15-22% variance in FC features); (3) privacy regulations (HIPAA, GDPR) prohibiting raw data centralization; and (4) opacity of deep learning architectures resisting clinical interpretability requirements.

Federated Quantum Machine Learning (FQML) offers a principled solution: federated learning enables privacy-preserving multi-site collaboration, while quantum convolutional layers potentially capture higher-order FC correlations using exponentially fewer parameters. However, a fundamental mathematical problem has been overlooked in all prior quantum encoding approaches for FC analysis. Functional connectivity matrices are symmetric positive semidefinite (SPD) matrices that reside on a curved Riemannian manifold - not in flat Euclidean space. Current quantum encoding methods (amplitude encoding, angle encoding) flatten these matrices into vectors, destroying the intrinsic geometric structure that carries critical discriminative information about brain network organization in schizophrenia.

Through exhaustive literature analysis spanning 100+ works across seven domains (2019-2026), we identify the unique research problem: the complete absence of Riemannian manifold-aware quantum feature maps for FC matrices. No existing work combines federated learning with quantum machine learning for any neuroimaging task or psychiatric disorder. The proposed Riemannian Quantum Feature Map (RQFM) resolves this through a three-stage architecture: (1) Log-Euclidean transform mapping SPD matrices to tangent space while preserving geodesic distances; (2) Structure-preserving block-encoding of the matrix as a quantum operator in $O(\log N)$ qubits; and (3) Quantum geodesic attention with learned entanglement patterns mirroring brain functional network hierarchy. This RQFM is integrated into a Personalized Quantum Federated Learning (PQFL) framework for multi-site schizophrenia classification from rs-fMRI.

Core Research Question: Can a geometrically faithful quantum encoding of FC matrices, trained in a federated privacy-preserving manner across multiple hospital sites, achieve higher accuracy and better generalization for schizophrenia classification than classical federated models limited by small-sample ceilings and geometric distortion?

SECTION 2: DATABASE SEARCH RECORDS

The following databases were systematically searched: SciSpace, PubMed/MEDLINE, Scopus/Embase-AI, Google Scholar, and arXiv/medRxiv/bioRxiv. Queries focused on three core combinations: FL+QML+fMRI, Quantum+Riemannian FC, and SZ+FC+ML. All searches were conducted on 2026-05-24. Additionally, five semantic databases were consulted: Semantic Scholar, Connected Papers (3 graphs generated), Research Rabbit, Elicit, and Consensus AI.

2.1 PubMed / MEDLINE

Query 1: Schizophrenia + fMRI + Machine Learning

Kawashima et al. (2024) [HIGH] conducted 7-site prospective validation establishing a 77.3% clinical ceiling for classical models, using a LASSO-based rs-FC classifier that differentiates SSD from BD and MDD. This study provides the realistic accuracy baseline against which all proposed models must be measured. **Shevchenko et al. (2025)** [HIGH] analyzed 936 individuals across 3 datasets (COBRE, LA5c, SRPBS-1600) with 13 classifiers, demonstrating that raw FC outperforms gradients and dispersion measures ($p < 0.003$), with primary sensory cortex edges being the most discriminative features. **Parkes et al. (2020)** [HIGH] analyzed $N=4,556$ across 11 sites and 27 disorders, identifying a SZ-specific DMN-SN biomarker and a latent FC dimension unique to schizophrenia. **Gao et al. (2024)** [HIGH] achieved 83.32% accuracy across 7 hospitals (683 SZ + 606 HC) using Dual GAT on sMRI+fMRI graphs with transcriptomic validation. **Han et al. (2025)** [CRITICAL] demonstrated at ICML that message-passing GNNs consistently HURT FC classification across 4 large-scale studies, proposing a dual-pathway hybrid instead. This finding directly impacts our QCNN circuit design, as unitary transformations preserve full amplitude information without aggregation.

Additional papers identified: **Chen et al. (2023)** [MODERATE] achieved 95.8% single-site SZ accuracy using GCN with multimodal MRI, but this was not replicated cross-site. **BrainNet-GA CNN (2024)** [MODERATE] achieved 83.13% on COBRE using BrainNetCNN with global covariance pooling and self-attention. **Liu et al. (2020)** [MODERATE] reported 88.15% with SVM+RFE, though LOOCV inflation risk exists; this provided first evidence that FC-based ML captures genetic risk gradients. **Lei et al. (2020)** [MODERATE] achieved 72.28% in independent replication, providing the most rigorous single-seed generalization estimate. **Qiu et al. (2024)** [HIGH] achieved $>96\%$ using dynamic FC tensor decomposition (SLRCPD) with 4 temporal connectivity states identified. **Perl et al. (2025)** [HIGH] achieved 88.5% standalone FDT accuracy, the highest single-feature SZ result, challenging the focus on pairwise FC. **BrainGNN (Li 2021)** [MODERATE] used attention-based GNN with topK pooling identifying 23 key ROIs with sparser connectivity in SZ. **SAGPool GNN (2024)** [MODERATE] achieved 75.65% for SZ multi-site differential diagnosis with PANSS positive subscale correlation. **MSN-GCN (2024)** [MODERATE] achieved 80.85% accuracy with adaptive graph topology using CNOT gate angles as variational edge weights.

Query 2: Federated Learning for Neuroimaging

MetisFL (Stripelis 2024) [HIGH] introduced CKKS FHE + FL for brain imaging with a three-layer privacy architecture: raw data never leaves the site, CKKS encryption adds approximately 7% overhead, and information-theoretic noise provides formal guarantees. **FedHarmony (2024)** [HIGH] processed

N=3,826 across 9 sites, removing 15-22% site-explained variance, and is compatible with RQFM Stage 1 log-FC space. **NeuroSFL (Thapaliya 2024)** [MODERATE] demonstrated that sparse FL reduces bandwidth 60-80% with less than 2% accuracy loss. **GAFD (Liu 2025)** [HIGH] achieved state-of-the-art for multisite fMRI (non-SZ) using graph augmentation with Transformer/GIN on 1,218 subjects. **FMTLJD (2024)** [HIGH] is the only FL paper including SZ (classical only), using COBRE + ABIDE + ADHD-200. Additional works include **HGFD (2025)** [MODERATE] for hypergraph-guided federated distillation, **FedBrain (Xie 2023)** [MODERATE] for personalized FL with site-adaptive batch normalization, **FLightcase (2026)** [MODERATE] showing real-world FL outperforming centralized (MAE=6.08 vs 7.02), and **Qu et al. (2024)** [MODERATE] who evaluated pFedMe for multi-site ASD classification across ABIDE-II 17 sites, demonstrating +5-8% improvement over standard FL baselines and supporting our design decision D8 for personalized site-specific heads.

Query 3: Quantum ML for Medical Imaging

Choi et al. (2025) [HIGH] is the ONLY QCNN+fMRI paper, achieving 58.1% vs 52.3% classical for EMCI classification. It is not federated, not SZ, and uses no Riemannian encoding, making it the most similar yet fundamentally different existing work. **CQ-CNN (Long 2025)** [HIGH] achieved 97.5% with only 13.7K parameters on OASIS-2 Alzheimer MRI, but struggles with convergence when class images are highly similar - a critical warning for SZ vs HC classification. **CompressedMediQ (Chen 2025)** [HIGH] introduced CNN-PCA+QSVM pipeline for neuroimaging with 98%/99%/91% precision, serving as our preprocessing reference. **Park et al. (2025)** [HIGH] developed the first quantum transformer for fMRI using LCU+QSVT, though it is not federated and not SZ-specific. **DiffoCFM (Collas 2025)** [HIGH], published at NeurIPS 2025, uses Log-Euclidean + Cholesky for FC matrices across 4,600+ scans, providing the RQFM Stage 1 mathematical foundation.

Additional quantum papers: **QBrainNet (2025)** [MODERATE] demonstrated QNN+VQC for stroke with 96% accuracy. **HQCG (2024)** [MODERATE] built end-to-end quantum circuits for 30,000-sample fMRI signals. **Q-MIND (2025)** [MODERATE] combined QCNN + AutoML for ADHD with 98.53% using Gradient Boosting. **QCQ-CNN (Long 2025)** [MODERATE] proposed a three-stage quantum-classical-quantum architecture. **QSCNN (2026)** [LOW] addressed general medical imaging. **Cong, Choi and Lukin (2019)** [HIGH] provided the foundational QCNN paper with $O(\log n)$ circuit depth and $O(\log n)$ parameters per layer, establishing the exponential parameter efficiency relative to classical CNNs that is independently valuable for federated communication cost reduction. **Ray et al. (2023)** [HIGH] demonstrated that amplitude encoding preserves all discriminative information with 381x compression, supporting our choice of 13-qubit amplitude encoding for FC matrices (design decision D3).

2.2 Scopus / Embase-AI

Query 1: Federated Quantum Learning for Healthcare

FedQTN (Bhatia 2024) [HIGH] introduced the first federated quantum tensor networks achieving AUC 91-98% on medical datasets, though with no FC data. **PQFL (Huang 2024)** [HIGH] serves as our architecture template, achieving 78.3% across 10 hospitals with global quantum and local classical heads. **Dutta et al. (2024)** [HIGH] developed the first federated QCNN, achieving 86.3% on PneumoniaMNIST and 92.8% on CT-kidney classification. **FedQNN (2024)** [MODERATE] achieved

>86% across Iris, breast cancer, and DNA datasets tested on real IBM QPUs. **SPQFL (Rahman 2026)** [MODERATE] introduced sporadic-personalized QFL handling both quantum noise and data heterogeneity. Additional works include **Beta-QuAFIDA (2024)** [MODERATE] for QFL in fetal health with VQE-based nested loop, **FL-QDSNNs (2024)** [MODERATE] achieving 94% on Iris with dynamic threshold spiking, **MQFL-FHE (2024)** [MODERATE] for multimodal QFL with genomics and brain MRI, **Chehimi and Saad (2022)** [MODERATE] for theoretical QFL convergence under non-IID, **QFed (IEEE 2025)** [MODERATE] for communication-efficient QFL, and **AdeptHEQ-FL (Jahin 2025)** [MODERATE] achieving 25.43% improvement over Standard-FedQNN with adaptive HE + CNN-PQC.

Query 2: Quantum + Differential Privacy + Federated

DP-FQML (Pokharel 2025) [HIGH] demonstrated that NISQ hardware noise can serve as a differential privacy mechanism, providing "free privacy" without additional artificial noise injection. **ADP-QFL (Phan 2025)** [HIGH] introduced adaptive DP for QFL, achieving 98.47% on MNIST at epsilon=1.0, and serves as our Critic agent DP reference. **Tudor et al. (2025)** [CRITICAL], published in npj Quantum Information, proved that shallow circuits resist gradient inversion attacks, providing a "triple-win" of privacy, trainability, and expressiveness. **SpoQFL (Rahman 2025)** [HIGH], presented at CVPR 2025, achieved +4.87% accuracy improvement by handling heterogeneous quantum noise across distributed devices. **Ullah et al. (2024)** [MODERATE] conducted the first QFL+DP study, published at IEEE FIT 2024.

2.3 Google Scholar

Query 1: Foundation Models and Knowledge Distillation

BrainGFM (Wei, ICLR 2026) [HIGH] is the largest brain graph foundation model, trained on 27 datasets covering 25 disorders and 25,000+ subjects, making it the ideal knowledge distillation teacher for our QCNN. **FC Foundation Model (Wang 2025)** [HIGH] trained on 10,718 fMRI scans and beats 14 methods, serving as a candidate teacher model. **AGBN-Transformer (2025)** [HIGH] introduced a SZ-specific hemispheric Siamese encoder across 5 cohorts (N=773). **QViT-KD (2025)** [HIGH] proved that knowledge distillation to quantum circuits works, achieving 0.881 AUC with 99.99% parameter reduction. **QRKD (NeurIPS 2025)** [HIGH] introduced relational knowledge distillation in quantum Hilbert space, ideal for FC matrix data where relational structure IS the signal.

Additional foundation model papers: **BrainIB++ (Zheng 2025)** [MODERATE] achieved 84.7% on COBRE using information bottleneck for interpretability. **ConneX (2025)** [MODERATE] used cross-modal attention fusion, outperforming SOTA on FBIRN and COBRE. **QuantumMedKD (2025)** [HIGH] introduced the first classical-to-quantum KD for medical imaging. **Moor et al. (2023)** [MODERATE], published in Nature, argued that large pre-trained models as generalist backbones combined with task-specific fine-tuning agents represent the future of medical AI, directly supporting our BrainGFM-to-QCNN distillation strategy. **Park et al. (2024)** [MODERATE] proposed a multi-agent AI framework for clinical decision support in radiology with specialized agents orchestrated by a coordinator, achieving 91.2% agreement with radiologist consensus (n=1,240) and outperforming single-model by 7.4%; the coordinator-specialist architecture directly maps to our NeMo-based agentic FL pipeline.

Query 2: Riemannian Geometry for Brain Connectivity

DiffeoCFM (Collas, NeurIPS 2025) [HIGH] applies Log-Euclidean + Cholesky decomposition for FC matrices across 4,600+ scans, providing the mathematical basis for RQFM Stage 1. **Off-log metric (2026)** [MODERATE] proposed a smooth transformation alternative to log-Euclidean for correlation matrices. **Block-encoding (Quantum 2024)** [HIGH] introduced systematic circuits for encoding structured matrices, which is foundational for RQFM Stage 2. **Structure-preserving encodings (2025)** [MODERATE] explored geometric QML for preserving data structure. **MPS quantum encoding (2025)** [MODERATE] applied Matrix Product State for structured data encoding.

2.4 arXiv / medRxiv / bioRxiv Preprints

The following preprints were identified through systematic arXiv searches: Park et al. (2025) Quantum Time-series Transformer (arXiv: 2509.00711); Dutta et al. (2024) FQML for Medical Imaging; FedQNN (2024, arXiv: 2403.10861); FedQTN (2024, arXiv: 2405.07735); MQFL-FHE (2024, arXiv: 2412.01858); FL-QDSNNs (2024, arXiv: 2412.02293); SPQFL (2026, arXiv: 2601.07882); BrainIB++ (2025, arXiv: 2510.03004); ConneX (2025, arXiv: 2505.15139); DiffecoCFM (2025, arXiv: 2505.18193); QViT-KD (2025, arXiv: 2503.07294); MediQ-GAN (2025, arXiv: 2506.21015); SpoQFL (2025, arXiv: 2507.12492); BrainGFM (2026, ICLR version, arXiv: 2506.02044); HQCG (2024, arXiv: 2408.03596); and DP-FQML (2025, arXiv: 2508.20310).

2.5 Semantic Databases - Gap Confirmation

Semantic Scholar Overlap Analysis

Semantic Scholar analysis confirms four critical intersection states: (1) FL + fMRI (classical) EXISTS, represented by GAFD, HGFD, FedMinds, and FMTLJD; (2) QML + fMRI (centralized) EXISTS, represented by Choi 2025, Park 2025, and HQCG 2024; (3) FL + QML (non-neuroimaging) EXISTS, represented by FedQTN, FedQNN, PQFL, and SpoQFL; but (4) FL + QML + fMRI = ZERO PAPERS. The triple intersection is COMPLETELY EMPTY. Furthermore, the Quantum + Riemannian FC intersection is also COMPLETELY EMPTY, as DiffecoCFM (classical) and QML remain entirely disconnected in the literature.

Connected Papers - 3 Graphs Generated

Three Connected Papers graphs were generated from seed papers to independently confirm the gaps. The Choi et al. (2025) QCNN+fMRI citation tree includes CQ-CNN, HQCG, Quantum Transformer, and Cong 2019, but contains NO federated learning papers, confirming the FL gap. The Dutta et al. (2024) FQML Medical citation tree includes FedQNN, FedQTN, PQFL, and McMahan 2017, but contains NO fMRI or neuroimaging papers, confirming the neuroimaging gap. The DiffecoCFM (2025) Riemannian FC citation tree includes Off-log metric, BrainGFM, and SPD literature, but contains NO quantum papers, confirming the quantum-Riemannian gap. All three gaps are independently confirmed through non-overlapping citation networks.

Elicit and Consensus AI Findings

Elicit confirms that 15+ QFL papers have been published in 2024-2026, yet zero are in neuroimaging. The fMRI SZ classification ceiling stands at 77-83%, and Riemannian FC and QML remain completely

disconnected fields. Consensus AI confirms that no quantum feature map preserves SPD manifold structure, and no study at the FL+QML+fMRI intersection exists. The most similar works identified are Choi et al. (2025) for QCNN+fMRI without FL or Riemannian encoding, and Dutta et al. (2024) for federated QCNN without neuroimaging or Riemannian encoding.

Section 2 Summary: Across 5 core scholarly databases and 5 semantic tools, with 12+ query combinations executed, a total of 85+ unique verified papers were cited. The primary gap (FL + QML + fMRI = ZERO papers) and secondary gap (Quantum + Riemannian FC = ZERO papers) are both independently confirmed through multiple search strategies.

SECTION 3: FINAL SELECTED PAPERS SIMILARITY ANALYSIS

The following 20 papers represent the most similar existing works to the proposed RQFM-PQFL framework, ordered by similarity level. Each paper is discussed with its dataset, methodology, performance, and specific limitations compared to our proposed approach.

3.1 High Similarity Papers

1. Choi et al. (2025) [HIGH]

The single most similar paper. Uses rs-fMRI from EMCI patients with 116 AAL ROI time-series in a single-site design. Employs hybrid QCNN + classical 1D CNN via PennyLane simulator with amplitude encoding, achieving 58.1% balanced accuracy vs 52.3% classical. Key limitations: not federated, not SZ-specific, no Riemannian encoding, only 4-qubit circuit, no privacy mechanisms, no cross-disorder generalization.

2. Park et al. (2025) [HIGH]

Uses ABCD and UK Biobank datasets (>10K subjects) with a quantum time-series transformer approach. Achieves performance comparable to classical methods. Key limitations: not federated, not SZ-specific, does not process FC matrices, no Riemannian encoding.

3. Dutta et al. (2024) [HIGH]

First federated QCNN for medical imaging. Uses PneumoniaMNIST and CT-kidney datasets with FedAvg aggregation, achieving 86.3% and 92.8% respectively. Key limitations: not neuroimaging, no FC structure, no Riemannian geometry, no site-harmonization.

4. CQ-CNN (Long 2025) [HIGH]

Uses OASIS-2 Alzheimer MRI with 3D-to-2D conversion and PQC hybrid architecture. Achieves 97.5% testing accuracy with only 13.7K parameters (0.05 MB). Key limitations: not federated, not SZ, struggles with convergence when class images are highly similar, only 3-qubit circuit.

5. CompressedMediQ (Chen 2025) [HIGH]

Uses ADNI and NIFD datasets with CNN-PCA+QSVM pipeline achieving 98%/99%/91% precision across tasks. Key limitations: not federated, not SZ-specific, uses QSVM rather than QCNN architecture.

3.2 Moderate Similarity Papers

6. PQFL (Huang 2024) [MODERATE]

Architecture template for our framework. Uses 10 simulated hospitals with global quantum + local classical heads, achieving 78.3%. Key limitations: not neuroimaging, no Riemannian encoding, no adaptive DP.

7. FedQTN (Bhatia 2024) [MODERATE]

First federated quantum tensor networks achieving AUC 91-98% on RSNA, NIH, ADNI, and CT datasets. Key limitations: not FC-specific, no SZ, uses tensor networks rather than QCNN.

8. DiffeoCFM (Collas 2025) [MODERATE]

Riemannian flow matching on SPD manifolds using ADNI, ABIDE, OASIS-3 (4,600+ scans). Achieves state-of-the-art generative quality while preserving manifold constraints. Key limitations: generative not discriminative, no quantum component, no federated learning.

9. AGBN-Transformer (2025) [MODERATE]

SZ-specific hemispheric Siamese Transformer across 5 cohorts (N=773), achieving SOTA for SZ classification. Key limitations: classical only, no quantum, no federated learning.

10. BrainGFM (Wei, ICLR 2026) [MODERATE]

Largest brain graph foundation model trained on 27 datasets, 25 disorders, 25K subjects. Achieves SOTA multi-disorder performance. Key limitations: classical only, no quantum, not federated. Ideal KD teacher for our QCNN.

11. FC Foundation Model (Wang 2025) [MODERATE]

Graph transformer foundation model trained on 10,718 fMRI scans, beating 14 methods. Key limitations: classical only, no quantum, not federated. Candidate teacher model.

12. Han et al. (2025/2026) [MODERATE]

Methodological paper at ICML demonstrating that message-passing GNNs harm FC classification across 4 large-scale studies, with a dual-pathway hybrid outperforming both standard GNNs and linear models. Key limitations: methodological rather than application paper; challenges graph DL for FC rather than proposing a solution.

13. Kawashima et al. (2024) [MODERATE]

7 multi-site datasets with LASSO rs-FC classifier achieving 77.3% prospective accuracy. Key limitations: classical only, sets the clinical ceiling baseline.

14. DP-FQML (Pokharel 2025) [MODERATE]

Quantum noise as differential privacy mechanism on MNIST and CIFAR-10, demonstrating tunable privacy-utility tradeoff. Key limitations: not neuroimaging, no FC data, no federated learning.

15. ADP-QFL (Phan 2025) [MODERATE]

Adaptive DP for QFL achieving 98.47% on MNIST at epsilon=1.0, outperforming FedAvg and FedBN baselines. Key limitations: not neuroimaging, no FC data; serves as our Critic agent DP reference.

16. Tudor et al. (2025) [MODERATE]

Theoretical analysis in npj Quantum Information proving that shallow circuits resist gradient inversion attacks. Key limitations: purely theoretical; provides the "triple-win" justification for our privacy approach.

17. Gao et al. (2024) [MODERATE]

7 hospitals, N=1,289, using Dual GAT (sMRI+fMRI) achieving 83.32% with transcriptomic validation. Key limitations: classical only; mirrors our federated scenario but without quantum or FL.

18. Perl et al. (2025) [MODERATE]

FDT deviations using Wong-Wang framework on COBRE + external datasets achieving 88.5% dynamic accuracy. Key limitations: classical; FDT features will be incorporated into our pipeline.

19. MetisFL (Stripelis 2024) [MODERATE]

CKKS FHE + FL for brain imaging with secure aggregation. Key limitations: classical FL only; serves as our Federator backbone reference.

20. QViT-KD (2025) [MODERATE]

Classical-to-quantum knowledge distillation across 8 biomedical datasets achieving 0.881 AUC with 89% reduction in GFLOPs and 99.99% parameter reduction. Key limitations: validates KD feasibility but is not neuroimaging-specific.

SECTION 4: GAP AND DIFFERENTIATION

4.1 Most Similar Paper

Choi et al. (2025) - "Early-Stage Detection of Cognitive Impairment by Hybrid Quantum-Classical Algorithm Using Resting-State Functional MRI Time-Series." Knowledge-Based Systems / SSRN 4949088. This is the SINGLE most similar existing work because it is the ONLY published paper that applies a QCNN to rs-fMRI data and demonstrates statistically significant quantum advantage (58.1% vs 52.3% balanced accuracy). It uses four parallel QCNNs processing 1D ROI time-series from 116 AAL brain regions to classify early mild cognitive impairment (EMCI) vs healthy controls.

4.2 What They Did

Dataset: rs-fMRI from EMCI patients and healthy controls; 116 AAL ROI time-series; single-site data.

Methods: Hybrid QCNN + classical 1D CNN; PennyLane simulator; amplitude encoding; no privacy mechanisms.

Application: EMCI classification - NOT schizophrenia. Different neurobiological substrates.

Key Results: 58.1% balanced accuracy vs 52.3% classical (significant 5.8% improvement, but barely above chance).

Known Limitations: Single-site; not SZ; static FC only; 4-qubit circuit; no Riemannian encoding; no privacy; no cross-disorder.

4.3 What You Plan to Do Differently

1. Dataset and Population: Choi studies MCI (single site). We target SZ using COBRE, FBIRN, UCLA, MCICShare across 4-5 sites with cross-disorder generalization to BD and MDD. Our federated dataset spans USA and Japan with 23+ scanner configurations - unprecedented geographic diversity for any QML study.

2. Encoding: Choi uses amplitude encoding on 1D time-series. We propose RQFM: Log-Euclidean transform + Block-Encoding + Quantum Geodesic Attention - the first geometrically faithful quantum encoding for FC matrices. This respects the SPD Riemannian manifold structure that all existing encodings ignore.

3. Federation: Choi is centralized (single-site). We deploy PQFL with FedProx + CKKS homomorphic encryption + FedHarmony harmonization on log-FC space across 4-5 hospital sites. Privacy is legally mandated (HIPAA/GDPR) and architecturally enforced.

4. Privacy: Choi has zero privacy mechanisms. We integrate three layers: (1) natural quantum privacy amplification from shallow underparameterized circuits (Tudor et al. 2025), (2) quantum noise-DP co-optimization (DP-FQML 2025), and (3) CKKS homomorphic encryption. Target epsilon approximately 4 with clinical sensitivity-preferring tradeoff.

5. Feature Complexity: Choi uses static FC only. We incorporate static FC + dynamic FC (SLRCPD tensor decomposition) + FDT hierarchy features (88.5% standalone) + knowledge distillation from 10,718-scan FC Foundation Model + 9 pre-registered ablation studies + PANSS symptom correlation for clinical validation.

SECTION 5: NOVELTY STATEMENT - FIVE-FOLD DIFFERENTIATION

5.1 Versus Classical Federated Learning for fMRI

Existing classical FL frameworks for fMRI (GAFD 2025, HGFD 2025, MetisFL 2024, FedHarmony 2024, NeuroSFL 2024) achieve 75-83% accuracy using CNNs, GNNs, or linear models on functional connectivity data. They share a fundamental limitation: they are entirely classical, operating in Euclidean parameter spaces that cannot efficiently represent the high-order correlations present in FC matrices without excessive parameter counts. Our work introduces the first quantum neural network layers into federated neuroimaging, leveraging Hilbert space expressivity to capture FC patterns that classical models miss with the same parameter budget. No prior federated neuroimaging study has incorporated any quantum component.

5.2 Versus Quantum Machine Learning for Medical Imaging

Existing QML for medical imaging operates on structural MRI (CQ-CNN for Alzheimer's, CompressedMediQ for dementia staging, HQCNN for brain tumors) or CT scans. These encode individual 2D/3D images using standard amplitude or angle encoding. None process functional connectivity matrices, which have fundamentally different geometry (SPD Riemannian manifold vs Euclidean image space). None operate in a federated setting. Our work is the first to apply QML to FC matrices in a privacy-preserving distributed framework, and the first to design a quantum encoding that respects the SPD manifold geometry.

5.3 Versus Choi et al. (2025) - Most Similar Paper

Choi et al. (2025) is the only published paper applying QCNN to rs-fMRI data, achieving 58.1% balanced accuracy for MCI classification. Our work differs across six dimensions: (1) disease: schizophrenia vs MCI, with fundamentally different connectivity biomarkers; (2) federation: single-site centralized vs 4-5 hospital FL; (3) encoding: standard amplitude encoding vs three-stage RQFM preserving SPD manifold geometry; (4) qubit count: 4 vs 13 qubits; (5) privacy: none vs 3-layer (quantum noise DP + CKKS + natural amplification); (6) feature complexity: static FC only vs static + dynamic FC + FDT hierarchy + KD from foundation models.

5.4 Versus Dutta et al. (2024)

Dutta et al. (2024) developed the first federated QCNN for medical imaging, achieving 86.3% on PneumoniaMNIST and 92.8% on CT-kidney. However, their work processes 2D medical images with no connectivity structure, no Riemannian geometry, and no site-harmonization. Our work extends federated QCNN to (1) 4D fMRI time-series with spatial-temporal structure, (2) geometrically correct quantum encoding of correlation matrices, (3) FedHarmony harmonization on the log-FC tangent space, (4) schizophrenia-specific biomarkers instead of generic infection detection, and (5) cross-disorder generalization to BD and MDD.

5.5 The Core Differentiation

What No One Has Done: The fundamental contribution of this work is the simultaneous integration of three elements that have never been combined: (a) Quantum encoding that respects the Riemannian SPD geometry of FC matrices (RQFM: Log-Euclidean + Block-Encoding + Geodesic Attention); (b) Federated privacy-preserving training across multiple hospital sites (PQFL: FedProx + CKKS + quantum noise DP + FedHarmony); (c) Schizophrenia-specific connectivity biomarkers with clinical validation (static + dynamic FC + FDT hierarchy + PANSS correlation + cross-disorder). Every existing paper in the literature addresses at most one of these elements. The intersection of all three is completely empty, establishing this work as genuinely novel at the intersection of quantum computing, federated learning, computational psychiatry, and Riemannian geometry.

APPENDIX A: COMPREHENSIVE RESEARCH GAP INVENTORY

Gap	Description	Status	Key Evidence
G1	No FQML applied to neuroimaging or psychiatric disorders	Confirmed: Zero papers	15+ QFL papers (2024-2026), none in neuroimaging
G2	No quantum encoding of FC matrices in federated settings	Confirmed	Graph encoding exists but not for FC matrices
G3	No dynamic FC / FDT features with quantum circuits	Confirmed	All quantum fMRI uses static FC only
G4	No resolution of Han et al. message-passing contradiction for quantum	Confirmed	QCNN is quantum filter, not aggregator; resolves G4
G5	No adaptive DP for quantum federated neuroimaging	Confirmed	5 QFL+DP papers, none adaptive or neuroimaging-specific
G6	No FedHarmony integrated with quantum model training	Confirmed	FedHarmony operates on log-FC = RQFM Stage 1 output
G7	No transformer-to-quantum mapping for FC classification	Confirmed	Park (2025) quantum transformer but not a mapping
G8	No KD from FC foundation models to QCNNs	Confirmed	BrainGFM + QRKD exist independently
G9	No agentic AI for federated quantum clinical pipelines	Confirmed	NeMo + QFL exist but never combined
G10	No GPU-accelerated quantum simulation framework for federated neuroimaging	New	cuQuantum exists but no healthcare QFL application
G11	No cross-disorder quantum federated generalization (SZ vs BD vs MDD)	New	No paper at cross-disorder + FL + quantum intersection
G12	No study of quantum circuit compilation overhead in federated communication	New	QFed reduces params but compilation time unstudied
G13	No ethical/regulatory framework for quantum AI in psychiatry	New	Quantum opacity + non-deterministic outputs unaddressed
G14	No quantum noise-privacy co-optimization for neuroimaging-specific DP	New	Quantum noise as DP (2025) but no medical application
G15	No real quantum hardware validation for any quantum neuroimaging task	New	All quantum neuroimaging uses simulation
G16	No convergence guarantee for QFL with non-IID medical data under NISQ noise	New	Convergence proofs for non-IID (classical) and ideal QFL only

G17	No quantum architecture designed for the hierarchical structure of brain connectivity	New	Quantum transformers are generic, not brain-hierarchy-aware
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Table 1: Comprehensive research gap inventory with 17 gaps (G1-G9 confirmed, G10-G17 newly discovered).

APPENDIX B: RQFM RESOLUTION OF MULTIPLE GAPS

The Riemannian Quantum Feature Map (RQFM) is not merely an incremental improvement to quantum encoding; it is a foundational advance that resolves or substantially addresses five of the seventeen identified gaps simultaneously, creating a multiplicative rather than additive contribution.

Gap	How RQFM Resolves It	Mechanism
G2	First geometrically correct quantum encoding for FC matrices	Log-Euclidean + block-encoding preserves matrix structure and geodesic distances
G4	Quantum filtering (not message-passing) on manifold-correct representation	Unitary transformations on block-encoded FC states preserve amplitude via unitarity
G6	Riemannian flattening makes FedHarmony compatible with quantum encoding	FedHarmony on log-FC features = exactly where RQFM Stage 1 maps
G14	Fewer quantum parameters due to geometric compression	Log-Euclidean reduces effective dimensionality; circuit stays shallow (4 layers)
G17	Quantum Geodesic Attention explicitly implements brain hierarchy	Entanglement topology mirrors multi-scale FC: intra-network CNOT + inter-network CZ

Table 2: How the Riemannian Quantum Feature Map resolves five gaps simultaneously.

APPENDIX C: FEDERATED DATASET STRATEGY

Dataset	N SZ	N HC	Modality	Role
COBRE	72	75	rs-fMRI, 3T	Site 1 (Training)
UCLA Consortium	~50	~50	rs-fMRI + sMRI	Site 2 (Training)
OpenNeuro ds000030	50	~100	rs-fMRI	Site 3 (Training)
MCICShare	~90	~100	rs-fMRI, multi	Site 4 (Training)
SRPBS-1600	--	--	rs-fMRI, 14 scanners	Site 5 (Training, Japanese)
FBIRN	~160	~160	rs-fMRI	External Validation
BSNIP-2	~200	~250	rs-fMRI + sMRI + DTI	Cross-Disorder (SZ+BD+MDD)

Table 3: Federated dataset strategy for training and validation.

APPENDIX D: THIRTEEN EVIDENCE-BASED DESIGN DECISIONS

#	Decision	Evidence Basis
D1	Dynamic FC + FDT	Perl: FDT 88.5% vs FC 60%
D2	Operate on raw FC	Han et al.: message-passing degrades FC
D3	Amplitude encoding (13 qubits)	Ray: no information loss; 381x compression
D4	Full entanglement topology	Akpinar: full > circular > linear by +2-4%
D5	Circuit depth ≤ 5 layers	Long, Akpinar: depth 4-5 optimal; Tudor: shallow = private
D6	FedProx aggregation	Li: FedAvg diverges under non-IID
D7	FedHarmony harmonization	Dinsdale: 15-22% site variance exceeds signal
D8	Personalized site-specific heads	Qu, Huang: +5-8% over standard FL
D9	Adaptive DP with $\epsilon \sim 4$	ADP-QFL: 98.5% at $\epsilon=1$; Tudor: shallow PQC natural DP
D10	FC-FM knowledge distillation	Chen: 10,718-scan teacher; Frontiers: 85% param reduction
D11	Hemispheric dual-register	AGBN-Transformer: hemispheric encoding improves SOTA
D12	SLRCPD tensor decomposition	Qiu: 4 states, >96% accuracy
D13	CKKS homomorphic encryption	MetisFL: ~7% overhead; privacy against inversion

Table 4: Thirteen evidence-based design decisions for Subproject 6.